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DOCUMENT 5 — DISTRICT HEATING DELIVERY

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Component: District Heating Delivery System

Purpose:

Deliver useful heat to final consumers.

Inputs:

- Heat flow Q_{heat} [W]
- Boundary temperature $T_{\text{b_K}}$ [K]
- Environmental reference temperature $T_{\text{0_K}}$ [K]

Outputs:

- Delivered useful heat Q_{useful} [W]

Key Variables:

$T_{\text{b_K}}$: boundary temperature for heat exergy [K]

$T_{\text{0_K}}$: environmental reference temperature [K]

Equations:

$$\text{Ex_heat} = Q_{\text{heat}} * (1 - T_{\text{0_K}} / T_{\text{b_K}})$$

$$T_{\text{b_K}} = T_{\text{b_K}}$$

$$T_{\text{0_K}} = T_{\text{0_K}}$$

Assumptions:

- Steady-state heat transfer
- Network losses neglected